

SAMARTH PRATAP SINGH

AI Engineer | LLM Agents · RAG · Evaluation Harnesses · FastAPI/React

☎ +91-9452026413 ✉ samarthsin2006@gmail.com 🔗 [Linkedin](#) 🐙 [Github](#) 📁 [Portfolio](#) 🏆 [LeetCode](#) 🤗 [Hugging Face](#)

EDUCATION

VIT Bhopal University, Bhopal

2023 – 2027

B.Tech - Computer Science and Engineering - **CGPA - 8.61/10**

Bhopal, Madhya Pradesh

EXPERIENCE

AI/ML Engineering Intern

May 2026 – Present

AmberFlux EdgeAI Private Limited | Remote

Hyderabad, Telangana

- Built a PDF vision-extraction pipeline (GPT-5): cut processing from ~7 min to ~90 sec on 20-page drawings (~4.7×) via concurrent batch dispatch (`ThreadPoolExecutor` + `asyncio`), retries and schema-enforced outputs.
- Designed a cost-aware extraction flow (regex/Docling first, GPT-5 vision only for missing fields) for project name, builder, site, number, contractor, architect and dimensions from drawings, emails and text; values validated against internal project records at **>0.85** confidence on real lead sheets.
- Built a LangGraph-routed aggregation layer for documents up to **400 pages**, returning validated structured JSON with page-level guardrails and drawing-page exclusion.

PROJECTS

DoCopilot (Multi-Tenant RAG Platform) 🗄️ | [Next.js](#), [FastAPI](#), [PostgreSQL](#), [Qdrant](#), [Azure](#) May 2026

- Built a multi-tenant RAG platform with JWT auth, tenant-scoped retrieval via Qdrant payload filters, SHA-256 upload dedup, and an async ingestion job state machine (Celery + Redis via Docker Compose).
- Implemented hybrid retrieval (dense MiniLM + BM25 sparse, RRF) with cross-encoder reranking (top-20 → top-5) and SSE-streamed, source-cited answers; added regex-based input/output guardrails and per-IP rate limiting.
- Built an evaluation harness (40 questions, separate LLM judge: Llama-3.1-8B judging Qwen3-32B) to compare chunking and retrieval variants: **89.2%** correctness, **90.5%** relevance, **2.86s** avg latency; shipped via GitHub Actions (pytest CI → Docker → ACR → Azure Container Apps).

Argus - Multi-Agent Research Engine 🤖 🗄️ | [FastAPI](#), [LangGraph](#), [Docker](#), [LangSmith](#)

Feb 2026

- Built a LangGraph supervisor pipeline (planner, researcher, critic, writer) with LLM-driven `Command(goto)` routing over Tavily, ArXiv and Wikipedia, producing cited reports at **3** depth tiers (~20/45/90s).
- Isolated the human-in-the-loop interrupt in a dedicated node so routing LLM calls are not repeated when a paused run resumes; capped at **3** critique loops with a 30-min auto-finalize.
- Exposed a FastAPI submit → SSE-stream → fetch API with SQLite/Postgres auto-switching persistence and checkpointing, per-IP rate limiting, LangSmith tracing and Docker packaging.

ContextCore (Stateful Memory Agent) 🗄️ | [LangGraph](#), [PostgreSQL](#), [Qdrant](#), [MCP](#)

Jun 2026

- Built a LangGraph CLI agent with an intent router (task / memory-recall / chat) and conditional edges, executing task and note workflows through a custom FastMCP tool server (create/list/update/delete tasks, save/search notes) backed by PostgreSQL.
- Designed three memory layers (PostgreSQL checkpointing, Qdrant semantic recall, MongoDB user profile) with token streaming via `astream_events` and an async background memory writer so the prompt returns immediately.
- Wrote an LLM-judge eval harness (50 cases): **92%** pass rate; failures traced to tool-call and token-limit errors, fixed via tool-schema hardening.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL

ML/DL: PyTorch, TensorFlow/Keras, Hugging Face Transformers, PEFT/LoRA, Scikit-learn, NumPy, Pandas

LLM & Agents: LangGraph, LangChain, RAG, Tool-Calling, MCP, Prompt Engineering

Retrieval: Qdrant, FAISS, Hybrid Search (BM25 + Dense), RRF, Reranking

Backend & Web: FastAPI, React (Next.js), REST, SSE, Celery, Redis, PostgreSQL

Delivery: Docker, Azure Container Apps, GitHub Actions CI/CD, pytest, LangSmith

Evaluation: LLM-as-Judge, ROUGE, BERTScore, METEOR